

### Final Project Progress Report

After the initial proposal feedback, I have revised my project to consist of a few focused questions. The key questions of this analysis are:

- How have the topics of TED Talks changed over time based on their transcripts?
- Does LDA or BERTopic provide more interpretable topics of TED Talk transcripts?

The popularity analysis is no longer a core aspect of this project, but it is a potential follow-up analysis that can benefit from utilizing the work completed in this project as a starting point.

I have performed some exploratory data analysis. This dataset has 4005 TED talks. There were multiple dates available for each talk. I chose to use recorded date over publication date as a talk could be published years after being recorded. For example, some of the talk recordings are published as part of the TED Archive initiative which saw old talks uploaded to the website. This is the case for all the talks in the dataset that were recorded in the 1980s and 1990s. For the sake of this analysis, I decided that the recorded date was a better reflection of the time period of the talk as that is when the talk was written and given. The earliest recorded TED Talk in this data is from 1984 and the latest is from 2020. For my analysis I decided to group the low frequency years so that talks prior to 2009 are a single group, and talks from 2020 (incomplete year) are grouped with 2019. This ensured that each group had over 200 talks allowing for more robust analysis.

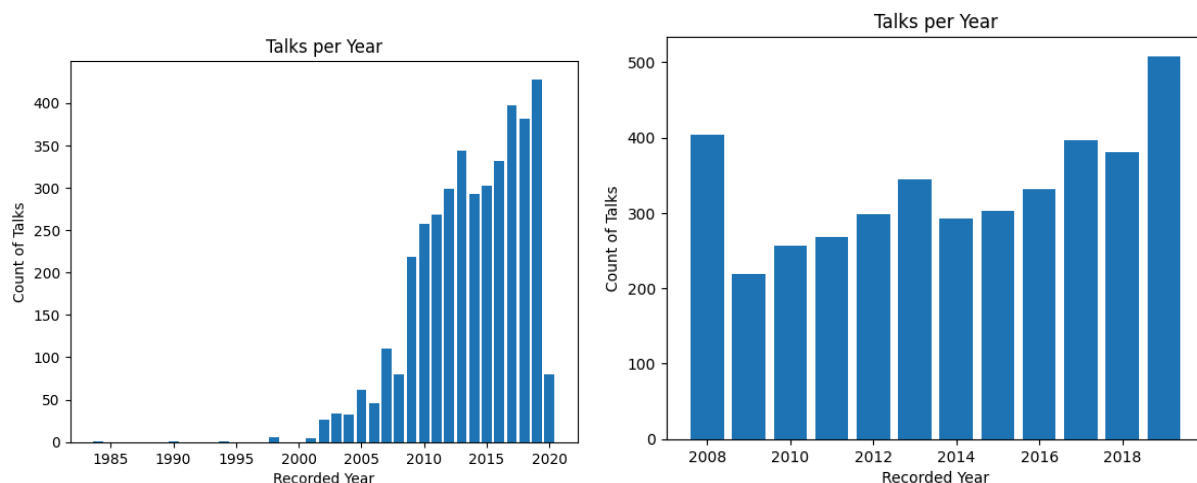


Figure 1. Histograms of talks per year before and after grouping

Each talk also has a list of TED assigned topics. There are 457 different topics which indicates that these topics are likely more granular than desired for this analysis. Within these topics, the most frequent are “science”, “technology”, and “culture”. I also looked at the most

frequent TED assigned topics per year to get insight into what to potentially expect in the planned temporal topic analysis. An observation was the high frequency of “society” and “social change” topics starting at about 2016. Prior to that the most frequent topics were often “science” and “technology”.

To measure topic change over time, a single topic model is fit to the entire corpus. Each TED Talk is then assigned a topic distribution, and average topic prevalence is compared across recording-year groups.

I started by training LDA topic models on the transcript texts. Topic coherence was used to compare candidate models because it measures the semantic consistency of the words within each topic. Initial experimentation showed highest coherence at 20 topics, the largest jump in coherence was from 5 to 10 topics.

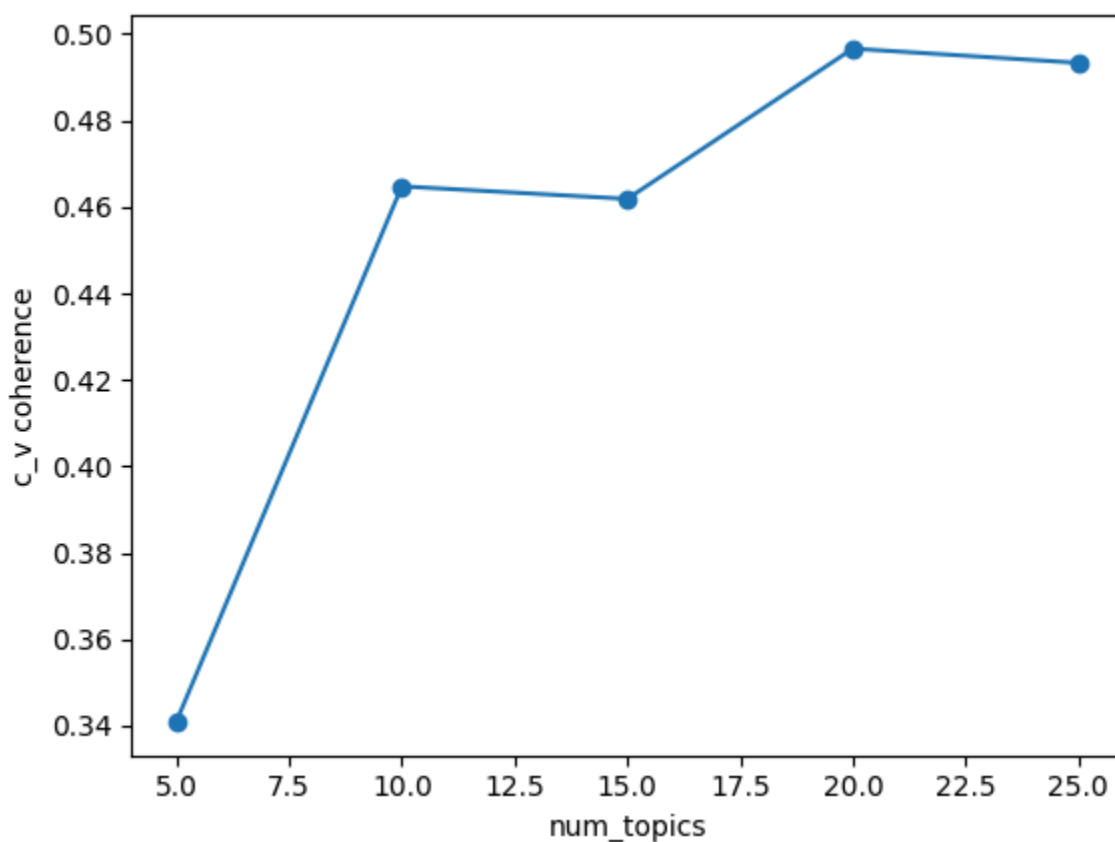


Figure 2. Overall LDA coherence from 5 to 25 topics

I decided to experiment with both 10 topics and 20 topics to see which topics were most interpretable.

In the 10 topic model, the individual topic coherence ranges from 0.209 to 0.565. The most coherent topic had the highest probability words “crime”, “civil”, “law”, “drug”, and “truth”. This is an intuitive topic about the criminal justice system. The least coherent topic had the top words “sea”, “percent”, “text”, “language”, and “crisis”. This topic appears less semantically coherent, suggesting it may combine several related but distinct themes.

For each recording-year group, I computed the mean topic probability across all talks and visualized these values as a heatmap. I also created a z-score normalized heatmap to emphasize temporal changes relative to each topic's average prevalence.

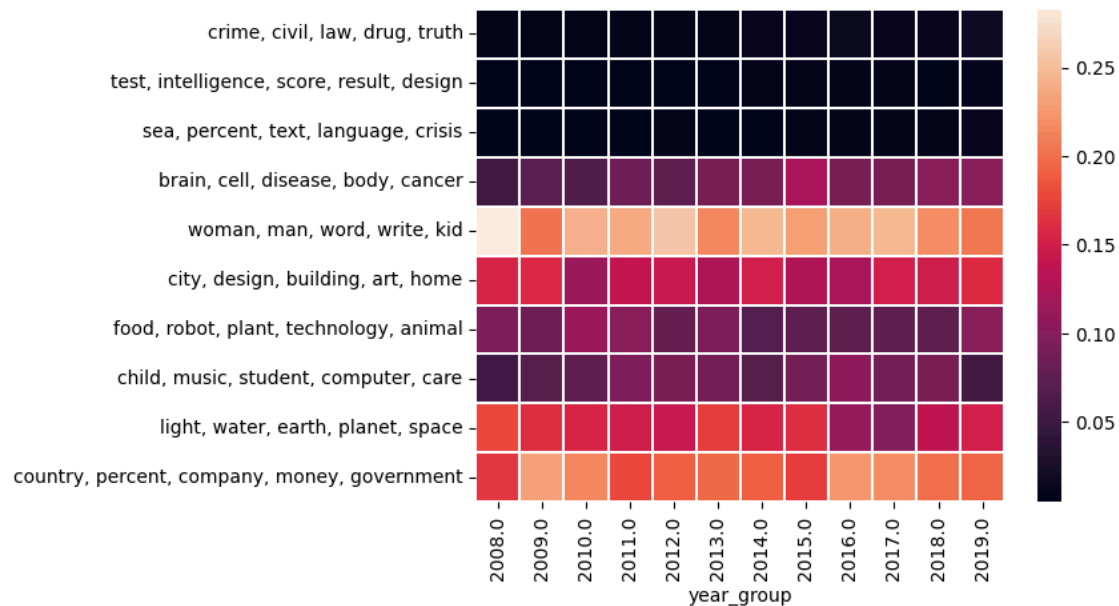


Figure 3. LDA topic mean probability heatmap with 10 topics

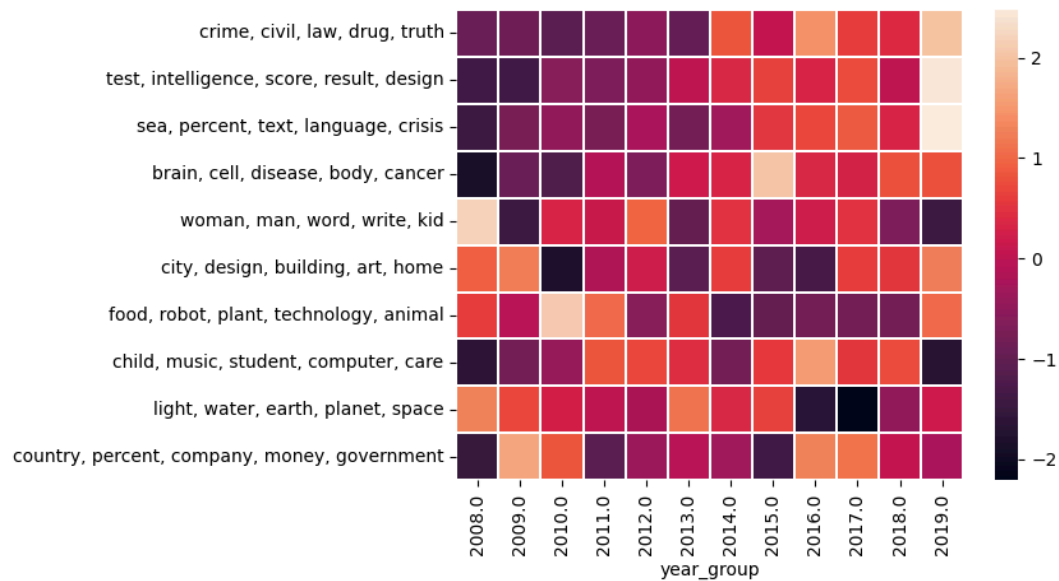


Figure 4. LDA normalized probability heatmap with 10 topics

The 20 topic model had individual topic coherence in the range of 0.280 to 0.722. Overall, there are a larger proportion of topics with coherence over .5 in the 20 topic model than the 10 topic model which indicates along with the overall coherence that 20 is a more fitting number of topics. The topic with the highest coherence has the top words “virus”, “vaccine”, “infect”, and “spread”. This is intuitively a topic about epidemiology. The least coherent topic

appears to be similar to the least coherent topic in the 10 topic model with the highest probability words of “percent”, “text”, “language”, “crisis”, and “conversation”. Upon investigation, the talks with the highest probability of this topic have the titles: “Crisis support for the world, one text away”, “How data from a crisis text line is saving lives“, and “Texting that saves lives” which appears to be a coherent if niche topic of crisis text lines. This suggests that the lower coherence score may reflect a relatively specialized topic rather than a poorly defined one.

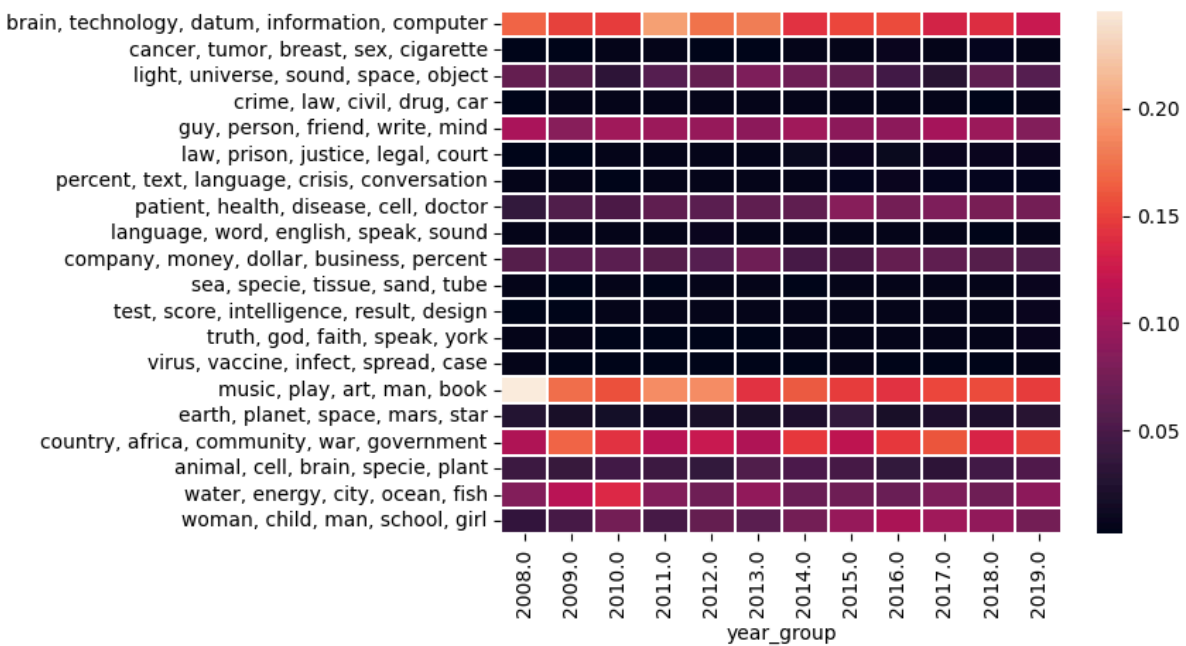


Figure 5. LDA topic mean probability heatmap with 20 topics

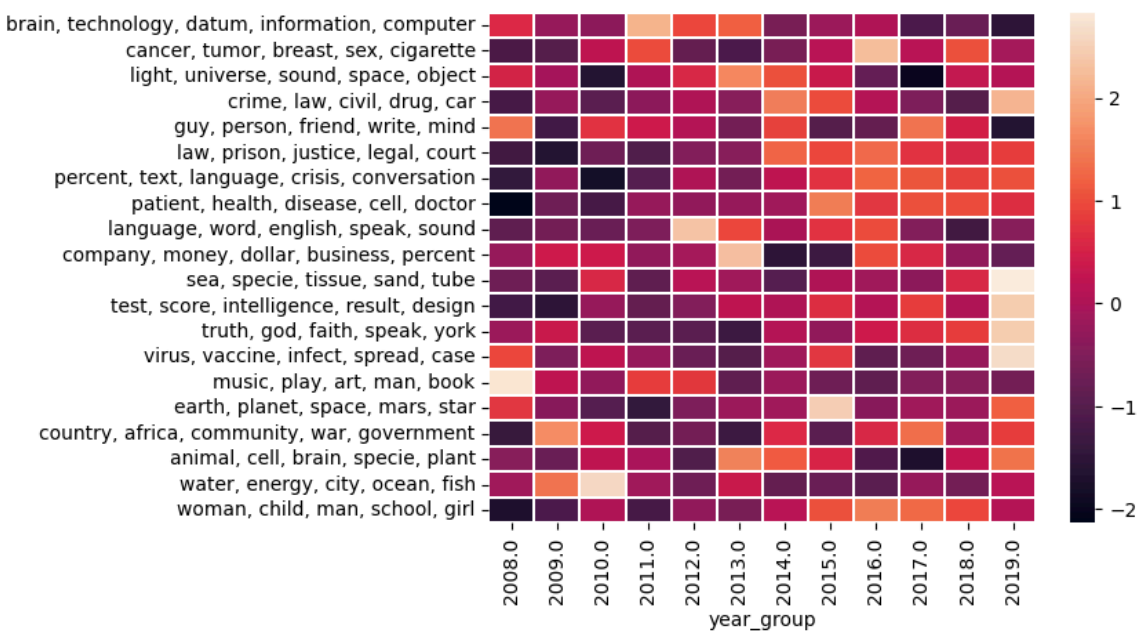


Figure 6. LDA normalized probability heatmap with 20 topics

My next step is to apply BERTopic, an embedding-based topic modeling approach, using the all-MiniLM-L6-v2 sentence transformer. I will experiment with the number of topics, starting with the inductively found number of topics from the model, then reducing to 20 and 10 topics for a more apt comparison to LDA. The comparison will consider both quantitative topic coherence and qualitative interpretability through manual inspection of representative topics and keywords. I will then choose which model I want to utilize for the rest of the temporal analysis and refine the topic labels to be more descriptive. The temporal analysis will consist of these heatmaps looking at both raw prevalence as well as Z-scores to identify trends. Finally, I will investigate rhetorical characteristics such as personal pronoun usage and motivational language over time to explore whether the rhetorical style of TED Talks has evolved alongside changes in topic prevalence.

#### References:

Meaney, C., Escobar, M., Stukel, T. A., Austin, P. C., & Jaakkimainen, L. (2022). Comparison of Methods for Estimating Temporal Topic Models From Primary Care Clinical Text Data: Retrospective Closed Cohort Study. *JMIR medical informatics*, 10(12), e40102.  
<https://doi.org/10.2196/40102>